

Reproducible Hybrid Recommendation for Vietnamese Retail

Abstract

Offline recommender results are difficult to interpret when model identity is reported without the data, split, candidate, masking, objective, and evaluator conditions that produced the result. This paper presents a provenance-aware protocol for assessing a decoupled Wide-and-Deep two-tower recommender in a controlled Vietnamese retail setting. The protocol fixes a temporal novel-purchase task, full-catalog ranking, seen-item masking, deterministic tie handling, per-user metrics, and a hierarchical uncertainty plan. It separates source-native reproduction, public protocol validation, controlled-benchmark comparison, and external validation into distinct evidence namespaces. The completed public lane contains one deterministic Pop run and three BPR runs on GroupLens MovieLens 100K; these rows are reported as descriptive protocol evidence only. The proposed Hybrid has not been admitted to the paper result set, and the controlled benchmark comparison remains a follow-up experiment. The contribution is therefore a reproducible comparison design with explicit evidence boundaries, rather than a conclusion about a settled model ordering.

Keywords: recommender systems, reproducibility, full-catalog evaluation, two-tower models, association rules, cold-item recommendation

1 Introduction

Retail recommendation is often presented as a model-selection problem, but an offline result is first a statement about a defined ranking task. A system must decide which user history is available, which items are eligible, what counts as a relevant future event, and how scores are converted into an ordered list. In large-scale services, candidate generation and ranking are also separate system roles rather than interchangeable names for one predictor [1] [2]. This distinction matters in retail settings with sparse user-item observations, changing purchase sequences, item metadata, and products with little or no training history. The same model label can describe different estimands when the candidate universe or target event changes.

The present study addresses that problem in a controlled Vietnamese retail setting. The benchmark is intended to represent the data structures needed for retail ranking, including user histories, product content, categories, price signals, and purchase or basket events. Its behavior is controlled and semi-synthetic rather than a direct sample of Vietnamese shoppers. That choice allows the protocol to expose specific mechanisms, such as rule-aligned co-purchase signals and item-side content, while limiting the claims that can be made about natural behavior. The benchmark declaration specifies 5,000 users, 5,200 items, 823,371 interactions, and a designated item-cold subset; these are manifest-level design facts and remain subject to the lineage receipt for the follow-up experiment.

1.1 Why comparability is a research problem

Offline recommender comparisons are sensitive to choices that are sometimes reported as implementation details. Temporal partitioning changes the information available at prediction time, while target construction determines which events enter the evaluation population. A recent study of splitting strategies demonstrates that sequential-recommender outcomes can vary with the split policy [3] and a systematic replicability analysis of BERT4Rec shows that training and implementation decisions are part of the reproducibility question [4]. Candidate construction, seen-item masking, negative sampling, cutoff, tie handling, and aggregation consequently define the quantity being estimated. Metrics with the same name are not automatically comparable when those inputs differ.

This paper responds by treating the evaluation protocol as a first-class study artifact. The protocol freezes an immutable temporal split, a complete item catalog, eligibility rules, relevance semantics, masking, deterministic ranking, and metric implementation. Each model exports scores to an independent shared evaluator. The evaluator, rather than the model implementation, applies masking, ranking, metric calculation, and per-user aggregation. This separation does not make every method equivalent. It makes the remaining differences inspectable and prevents a source-native number from silently becoming a result under a different task contract.

1.2 Distinct signals and the conditional Hybrid question

The candidate design combines a deep two-tower scorer with a rule-derived Wide component. The rationale is a complementarity hypothesis: a rule branch may retain explicit co-purchase regularities, while a content-aware deep branch may generalize beyond interactions that have already been observed. This hypothesis is related to, but not identical with, the Wide & Deep decomposition of linear cross-product memorization and embedding-based generalization [5] It is also distinct from item-based collaborative filtering, which aggregates neighbor evidence from interaction overlap [6] and from Bayesian Personalized Ranking, which specifies a pairwise objective for implicit feedback [7] The distinction is substantive: a change in encoder, objective, negative distribution, or system stage can change a result independently of the other components.

Association rules provide a mechanism vocabulary, not an automatic ranking claim. Apriori defines support and confidence as thresholds for mining frequent item relationships [8] and multi-item rule systems show how such relationships can be operationalized for recommendation under a particular dataset and evaluation design [9] Prior hybrid and next-basket studies provide useful component-comparison precedents, including combinations of preference, popularity, transition, and association signals [10] [11] [12] but those precedents do not determine the contribution of the proposed branch under the locked v5 cohort.

1.3 Research question and contributions

The primary research question is: under a fixed temporal novel-purchase full-catalog protocol on the controlled Vietnamese retail benchmark v5, does the proposed decoupled Wide-and-Deep Two-Tower Hybrid improve ranking effectiveness relative to the strongest faithfully reproduced baseline selected using validation data? Three subsidiary questions concern item-side cold performance, the incremental role of the rule branch, and replication on a compatible public commerce dataset. A fourth concerns efficiency, but it is opened only after the accuracy artifacts have passed their evidence gates.

The paper makes three contributions that are supported at this stage. First, it specifies a provenance-aware protocol in which data, split, candidate catalog, masking, evaluator, seeds, and model artifacts are explicit. Second, it separates source-native reproduction, public protocol validation, controlled-v5 comparison, and external validation into distinct evidence namespaces. Third, it reports an audited public protocol validation lane and preserves its limitations rather than using it to stand in for the unexecuted Hybrid comparison. The resulting claim is therefore about reproducible evaluation design and bounded evidence, not about a settled ordering of recommender architectures. The controlled Hybrid experiment remains a separately audited follow-up task.

The remainder of the paper first synthesizes prior work by signal and evaluation contract. It then defines the data and ranking protocol, describes the evidence lanes, reports the permitted public validation rows, and discusses the limits of the current evidence.

2 Related Work

2.1 Evaluation and reproducibility

Recommendation literature often compares model families through headline metrics, yet the metric is only interpretable together with the prediction event and candidate set. The split-sensitivity evidence in sequential recommendation makes temporal construction a substantive design choice [3] while the BERT4Rec replicability study identifies implementation and training choices that affect whether a reported result can be reproduced [4] These observations support a narrow methodological conclusion: a comparison should freeze

the target event, temporal boundary, candidate universe, masking, metric code, aggregation, and training budget before interpreting model differences.

The present protocol consequently distinguishes source-native evidence from harmonized evidence. A source-native result retains the source implementation, data preparation, split, candidate policy, and evaluator. A harmonized result uses a common data and evaluation contract only after the adapter’s deviations are declared. The distinction protects against a common category error in which a number reproduced from a paper is treated as if it had been generated under a new dataset or evaluator. It also makes a failed or incomparable reproduction a visible outcome rather than a reason to substitute a convenient surrogate.

The positioning is thus deliberately limited. Prior work motivates protocol control, but it does not supply a ranking for the v5 benchmark or authorize changing its frozen split after observing outcomes.

2.2 Collaborative, neural, and sequence recommendation

Item-based collaborative filtering estimates item similarity from historical co-interaction and aggregates neighboring items for a user [6] BPR instead defines a pairwise preference objective for implicit feedback and can be instantiated with matrix factorization or another differentiable scorer [7] These methods may share user and item identifiers, but they encode different assumptions about similarity, optimization, and unobserved interactions.

Neural collaborative filtering learns nonlinear interaction functions from user and item identifiers [13] DeepFM, by contrast, combines low- and high-order interactions over sparse feature fields for a click-through-rate task [14] The shared presence of embeddings does not make their input or target contracts interchangeable. This distinction informs the adapter registry: a model keeps its native objective and feature assumptions, while its score output is mapped to the common evaluator.

Sequential models add another source of non-equivalence. SASRec uses causal self-attention for next-item prediction [15] whereas BERT4Rec uses bidirectional sequence encoding and a masked-item Cloze objective [16] Their source evaluations use paper-native task and candidate designs. A next-item objective is therefore not identical to a next-basket set objective, and a source metric cannot be copied into the v5 table without a protocol bridge.

These families supply comparator and objective precedents, not directly comparable v5 results.

2.3 Wide and Deep, Two-Tower, and semantic representations

Wide & Deep jointly trains an explicit linear cross-product component for memorization with an embedding and multilayer component for generalization [5] That decomposition is useful for stating a mechanism hypothesis, but it does not identify the Wide component with association-rule mining. In the present design, the rule features are fit from the training transactions and exported as a separate score. The no-Wide condition is required to assess whether the branch contributes under the same evaluator.

Two-tower systems are frequently used in large-corpus retrieval. The YouTube system separates candidate generation and ranking [1] and neural retrieval work shows that negative-distribution and sampling-bias choices can be part of the model contract [2] DirectAU further illustrates that an objective can target alignment and uniformity rather than the pairwise objective used by BPR [17] The protocol must consequently identify whether a tower is a retriever, pre-ranker, or full-catalog ranker and must record its negative sampler.

Recent tower designs make the same boundary visible from another direction. ContextGNN introduces pair-aware local graph context while retaining an exploratory two-tower fallback [18] and T2Diff adds cross-interaction during training while targeting efficient matching [19] These works are useful controls for naming the pair-agnostic boundary. They do not imply an architecture ordering under the v5 full-catalog contract.

The positioning is therefore specific: this paper evaluates a decoupled architecture under one locked ranking contract and keeps encoder complexity, objective, sampling, and system stage as separate explanatory variables.

2.4 Apriori, association rules, and next-basket recommendation

Apriori defines a transaction-frequency support threshold and a conditional confidence threshold for mining association rules [8] Multi-item recommendation work demonstrates that such rules can be used to construct

a recommendation list, while its comparisons remain conditional on the source data and protocol [9] The distinction is important for the proposed Wide branch: a mined rule is a feature or score source, not evidence that the full ranking task has been solved.

Next-basket research also warns against treating all basket events as one behavioral regime. A reality-check analysis uses repeat ratios and baseline behavior to expose how aggregate accuracy can be shaped by repetition [20] and exploration-focused work isolates the harder question of recommending novel items [21] Mask-Swap and its BTBR baseline offer a direct next-novel-basket formulation, but their reported conditions do not warrant a universal conclusion across datasets or model families [22] Repeat/explore-aware graph work supplies further component and ablation precedent under its own data and protocol [23]

These studies motivate a train-defined rule-aligned cohort and a no-Wide ablation. They do not estimate the present mechanism question, and aggregate basket accuracy alone does not establish a novel-item gain.

2.5 Graph and contrastive recommendation

LightGCN simplifies graph convolution by propagating trainable user and item embeddings over observed interaction edges [24] Graph-contrastive controls such as SimGCL and LightGCL add contrastive or global-relation objectives within their own observed-graph settings [25] [26] An improvement on a sparse observed graph does not establish a representation for an item with zero training edges. The v5 protocol therefore reports sparse and strict zero-edge cohorts separately and does not infer cold-item capability from a graph score alone.

Graph results motivate comparator coverage and precise cold-item definitions, but they do not establish v5 performance or zero-edge performance.

2.6 Content, cold-item, and transfer learning

Cold-item analysis begins with the training graph. A strict zero-edge item has no training interaction; an item with one or more training edges may be rare, but it belongs to a different sparse cohort. Dropout-Net formulates cold-start training through an item-side framework [27] and ALDI defines its own cold-item distillation setting [28] Their thresholds and denominators cannot simply be transferred to v5. A reproducible cold-item analysis must construct cohorts before scoring and report warm, sparse, and zero-edge denominators separately.

Text encoders provide a useful item-side representation, but representation quality is not a ranking outcome. Sentence-BERT supplies a rationale for semantic sentence embeddings [29] whereas AlphaRec evaluates language representations inside a particular recommendation and zero-shot protocol [30] Sparse content-based cold-item work likewise defines its own problem and evaluation setting [31] The proposed pipeline therefore requires a recommender objective and a shared task-specific evaluator in addition to an embedding recipe.

Transferable item representations retain target-domain assumptions. UniSRec includes downstream adaptation [32] VQ-Rec evaluates vector-quantized transfer under its experimental setup [33] and UTGRec includes downstream fine-tuning [34] An architecture inspired by these works is not a replication unless the encoder, data lineage, adaptation procedure, split, candidates, metrics, and statistics are matched or every deviation is declared. Shared platform or benchmark lineage also means that pretraining and downstream results cannot be counted as independent replications merely because their domain labels differ [35] [31] Content aligned to collaborative teachers can also inherit popularity structure, so content availability alone does not guarantee an unbiased or effective cold-item ranking [35]

The positioning is conditional on full contract alignment. External and cold-item conclusions require matched encoder, data, adaptation, split, candidate set, metric, and statistical procedure; a source architecture name is not sufficient evidence.

3 Methodology and Protocol

3.1 Study design and estimand

The study is designed as a quantitative comparative offline benchmark. Its unit of analysis is a per-user metric vector nested within a training seed. This choice preserves the dependence created when two models are evaluated for the same users and permits uncertainty to be reported at both the user and seed levels. The estimand is comparative algorithmic performance under a declared dataset and evaluation protocol. It is not an estimate of changes in user behavior, revenue, satisfaction, or long-term production outcomes.

The central comparison is defined before the follow-up test is opened. The proposed model produces a score for every candidate item, and a locked baseline registry supplies the comparison score. The strongest baseline is selected from validation means using a fixed rule and is then frozen for the test evaluation. This sequence prevents test outcomes from determining the comparator, features, or stopping rule. A null or negative contrast remains a valid answer to the research question.

The protocol separates three operations that are often conflated. Training defines model parameters using only the training portion. Score export produces an ordered-catalog-compatible score artifact without calculating final metrics. Evaluation applies masking, ranking, relevance, and aggregation independently. This separation makes the score-producing implementation auditable and keeps metric semantics identical across model families.

3.2 Dataset and temporal protocol

The controlled v5 benchmark is a Vietnamese retail data construction with semi-synthetic behavior. Its manifest declares user histories, item content, category and price signals, and purchase or basket events. It also declares 5,000 users, 5,200 items, 823,371 interactions, and a designated item-cold subset. These values are treated as dataset-manifest facts until the fresh follow-up snapshot passes raw-to-internal reconciliation; they are not model outcomes.

The declared temporal boundaries are train from 2026-01-01 through 2026-06-19, validation from 2026-06-20 through 2026-07-10, and test from 2026-07-11 through 2026-08-01, all in UTC. User and item identifiers are mapped through an immutable raw-to-internal table. The adapter checks schema, duplicate rows, malformed records, timestamp ordering, user and item coverage, item-text provenance, and the availability of basket or co-purchase structure. An exclusion is recorded with its count and reason rather than silently removed.

The snapshot is admitted as a comparison input only when these checks form a continuous chain. The raw provider bytes, the normalized tables, the raw-to-internal identifier map, and the split manifest must each have a content hash. The split hash must resolve to the protocol version used by the runner, and the protocol version must resolve to the evaluator version used for the reported scores. This conjunctive rule matters because a correct model can still produce an uninterpretable result when, for example, an identifier map is regenerated after the split or item text is read from a later catalog revision. A failed link is recorded as an incomplete or diagnostic input; it is not repaired by silently substituting a nearby snapshot. Dataset suitability and model effectiveness are therefore assessed as separate questions. The former concerns whether the data can support the declared estimand, whereas the latter concerns the behavior of a model after the estimand has been fixed.

The public protocol validation uses GroupLens MovieLens 100K as a separate namespace and is not a substitute for the controlled v5 snapshot. It uses a locked RecBole 1.2.1 environment and a full-sort evaluation path. The public lane is useful for checking that a source-bound pipeline can materialize data, train a reference method, export metrics, and preserve configuration and checkpoint bindings. It does not establish that the same pipeline answers the Vietnamese retail research question.

3.3 Candidates, relevance, masking, and cohorts

For each eligible user u , the candidate set is the complete item catalog after removing items already observed in the allowed history:

$$C_u = I \setminus H_u^{\text{seen}}. \quad (1)$$

The catalog order is frozen by raw product identifier. Scores are ranked in descending order, with raw identifier ascending as the deterministic tie break. No sampled candidate set is used when full-catalog scoring is available. This policy makes the candidate universe explicit and avoids interpreting a sampled metric as an exact-catalog quantity.

The primary target is a novel organic purchase in the target split. A user is eligible only when at least one such positive event remains after the protocol filters. Binary relevance is used for the top- K metrics, with $K = 10$. For a user with relevant set T_u , the normalized discounted cumulative gain is

$$\text{NDCG@10}_u = \frac{\text{DCG@10}_u}{\text{IDCG@10}_u}, \quad (2)$$

where the ideal denominator uses $\min(|T_u|, 10)$. Hit rate is the indicator that at least one relevant item appears in the first ten positions. Recall is reported separately because its denominator is the number of relevant target items, not a hit indicator. Macro per-user GAUC is computed as the exact AUC between each user’s relevant positives and unseen non-positives, with average rank for ties, and then averaged across eligible users.

The item-cold analysis is defined from the training graph. A strict zero-edge item has no training interaction; an item with one or more training edges is reported, if needed, in a sparse cohort rather than the zero-edge cohort. Warm, sparse, and zero-edge denominators are constructed before scoring. The rule-aligned cohort is likewise constructed from training transactions only. These cohort definitions prevent a blended denominator from being presented as a strict cold-item result.

3.4 Models and adapter provenance

The registry begins with Random and MostPop sanity controls, followed by faithfully adapted reference methods where source and environment bindings are available. ItemCF and BPR represent different collaborative contracts: the former aggregates item-neighbor evidence, while the latter optimizes pairwise implicit preferences [6] [7]. The reference adapter retains the native objective and records repository revision, dataset mapping, split, sampler, configuration, checkpoint rule, and score-export revision. A source-native result is not relabeled as a harmonized result merely because its scores can be read by the shared evaluator.

The proposed scope contains three related conditions. **IndependentDeepTwoTower** uses user-history and item-content representations without the rule branch. **WideRuleOnly** uses association features fit from training transactions. **ProposedHybrid** combines the deep score and the rule score after the normalization and additive fusion specified in a frozen configuration. All three conditions emit component or total scores through the same **ScoreArtifact** contract. The association rules are fit before validation and test data are accessible, and the negative sampler draws only from items absent from the user’s training history.

The model descriptor also records the feature set, native objective, negative sampler, implementation provenance, configuration hash, adapter revision, and checkpoint hash. A text encoder is an input representation, not an outcome; recommendation quality is assessed only after the recommender objective and ranking evaluator have been applied. This constraint follows the distinction between semantic embedding quality and task-specific recommendation evaluation in prior work [29].

3.5 Metrics and statistical plan

NDCG@10 is the primary outcome. HR@10 and macro per-user GAUC are supporting outcomes, and Recall@10 is retained for reporting compatibility. Each run exports per-user vectors before aggregation. The final training seeds are 42, 2027, and 31415. A missing or failed required seed makes the confirmatory aggregate incomplete; it is not imputed or replaced after the test is opened.

For a model m and locked comparator b , the primary contrast is the mean of per-user differences across the three seed occurrences:

$$\Delta_{\text{NDCG}} = \text{mean}_{s,u} [\text{NDCG@10}_{m,s,u} - \text{NDCG@10}_{b,s,u}]. \quad (3)$$

The planned interval is a two-sided percentile 95% hierarchical paired bootstrap with 2,000 replicates. Each replicate resamples seed occurrences and, within each occurrence, resamples user indices with replacement.

The same user indices are used for the model and comparator. The aggregate analysis uses a NumPy PCG64 generator seeded with 42. No seed average is formed before the bootstrap.

The comparator, configuration, cohort builder, and evaluator are frozen before test access. Exploratory multi-model comparisons report the number of contrasts and apply Holm correction. Efficiency measures, including latency, throughput, and peak memory, are opened only after the accuracy artifact passes its required integrity checks. These choices keep statistical uncertainty separate from implementation reproducibility.

4 Experimental Design and Evidence Scope

4.1 Evidence namespaces

The paper uses four explicitly separated evidence surfaces. The first, `PUBLIC_DATA_PROTOCOL_VALIDATION`, contains the completed Pop/BPR MovieLens lane and supports descriptive statements about procedural execution and artifact traceability. The second, `OFFICIAL_PROTOCOL_REPRODUCTION`, is closed as incomparable and contributes no result row. The third, `HARMONIZED_V5_COMPARISON`, is sealed with major limitations and contributes no numeric row to this manuscript. The fourth is a future follow-up namespace for the v5 experiment and requires a new packet, registry, and audit.

This separation is substantive rather than administrative. A public run can validate an execution path while using a different target event, dataset, feature vocabulary, or source protocol. Conversely, a controlled benchmark can answer a tightly defined internal question without supporting a public-data generalization. The manuscript therefore does not pool metrics across these surfaces or use one surface to fill a missing row in another.

Admission is applied at the row level rather than to an entire project label. For a public row, admission requires a resolvable dataset binding, protocol binding, model configuration, checkpoint rule, evaluator receipt, and aggregate artifact. The four admitted rows satisfy that descriptive chain. They are retained because their provenance is replayable, not because their metric values are large or because they answer the primary Vietnamese retail question. The official source-native lane is retained as a separate incomparable outcome when its center, split, or evaluator cannot be aligned without changing the source contract. The harmonized v5 lane is likewise retained as a sealed limitation when mandatory comparator or statistical conditions are absent. These states are informative: they distinguish an execution path that was auditable from a scientific comparison that was not yet admissible. A later experiment may extend the evidence, but it must do so under a new namespace and a new admission decision rather than overwrite one of these states.

4.2 Run and artifact sequence

The follow-up execution begins with a dataset snapshot and protocol manifest. The runner then validates the environment, materializes the split, fits each model using its permitted training inputs, exports score chunks in the frozen catalog order, and invokes the independent evaluator. The resulting chain is *run manifest* to configuration and environment receipts, dataset binding, checkpoint manifest, score artifact, per-user metrics, aggregate metrics, and bootstrap report. Every link carries a SHA-256 binding or a replayable derivation from the preceding artifact.

The completed public lane follows the same evidence shape at a smaller scope. Its aggregate receipt is bound to SHA-256 512fb8ad8862f5e677ec18f369d310ae58c9d6f3c4f7b2e79c17ede389930c17. The lane contains one Pop run with seed 42 and three BPR runs with seeds 42, 2027, and 31415. The public result table is generated from that receipt and the four admitted row bindings; it is not reconstructed from console output.

4.3 Follow-up comparison and ablations

The controlled experiment first runs validation-only smoke checks and locks the comparator registry. It then trains the independent deep, rule-only, and full Hybrid conditions under the same data split, candidate catalog, masking, and evaluator. The full Hybrid can be interpreted only if its score artifact is finite, its catalog order matches the protocol, and its component artifacts are present.

Three seeds are run after the configuration and checkpoint rule are frozen. The test set is opened once, followed by per-user metrics and the hierarchical bootstrap. The rule mechanism is evaluated on the train-defined rule cohort, not on a cohort selected after seeing test results. The item-cold analysis reports strict zero-edge and sparse denominators separately. If the full model does not improve the locked contrast, the result is reported as null or negative; changing the metric or dataset after the observation is outside the protocol.

4.4 External validity and adaptation

An external dataset is admissible only when it preserves the essential task and input contract: persistent user meaning, purchase or basket relevance, item content required by the model, an auditable split, compatible candidate semantics, and an evaluator mapping. A session or next-interaction dataset may be useful for sensitivity analysis, but it is not automatically a purchase replication. Similarly, replacing a source encoder or adaptation procedure with an available text model changes the method contract. Transferable recommender architectures retain explicit assumptions about target-domain adaptation [32] [33] [34]. An external result is therefore labeled according to its actual alignment and deviations, not according to a shared retail application label.

5 Results

5.1 PUBLIC_DATA_PROTOCOL_VALIDATION

The admitted public lane completed four run rows: one deterministic Pop run and three BPR runs. Table 1 reports the five metrics retained in the aggregate receipt. The rows are descriptive protocol evidence; the lane has no paired per-user vectors or inferential test, and the single Pop run does not provide a seed-level variance estimate.

Table 1: Descriptive results in the PUBLIC_DATA_PROTOCOL_VALIDATION namespace.

Model	Seed	Recall@10	MRR@10	NDCG@10	Hit@10	Precision@10
Pop	42	0.042678	0.053217	0.031164	0.186837	0.021444
BPR	42	0.101423	0.105435	0.070481	0.283439	0.038535
BPR	2027	0.106237	0.113953	0.075893	0.299363	0.041507
BPR	31415	0.109880	0.116951	0.077937	0.301486	0.042569

For the primary descriptive metric, the three BPR values have mean 0.074770 and sample standard deviation 0.003853. The Pop point estimate is 0.031164. The BPR point estimates are consistent in direction across the three recorded seeds under this exact public protocol, but the design does not support an inferential conclusion because the Pop arm has one run and the lane lacks the paired per-user artifact required for such a comparison. The other metrics show the same row-level pattern, with BPR means of 0.294763 for Hit@10, 0.112113 for MRR@10, 0.105847 for Recall@10, and 0.040870 for Precision@10.

5.2 Evidence interpretation

The public rows demonstrate that the locked execution path can preserve a dataset binding, an environment receipt, a full-sort result, a checkpoint rule, and an immutable aggregate receipt across a small reference lane. They do not measure the proposed Hybrid, the v5 item-cold cohort, the rule-aligned cohort, or external commerce validity. The official reproduction lane remains incomparable, so no source-native target is imported into this table. The controlled v5 rows are likewise withheld from the result namespace used here.

The table should therefore be read as an audit sample of a protocol, not as a ranked leaderboard. Its four rows represent one deterministic control and three stochastic reference runs; they do not provide balanced replication across all methods or seeds. The BPR mean and dispersion summarize the recorded BPR runs only, while the Pop value is a single point observation. Neither summary changes the unit of analysis or

supplies a paired comparison. In particular, a row-level difference between the two labels would combine different run counts and would not be a valid estimate of the primary v5 contrast. Keeping this limitation beside the table prevents a descriptive sanity check from acquiring the rhetorical role of a scientific baseline.

6 Discussion and Limitations

6.1 What the current evidence supports

The main result of this stage is a protocol boundary. A recommender comparison becomes more interpretable when the prediction event, catalog, masking, relevance, evaluator, and training conditions are recorded together. The public lane provides a concrete, audited example of that boundary: four rows are traceable to a fixed environment, dataset binding, configuration, and checkpoint chain. The value of these rows is procedural. They show what was run and how the output can be traced, not what a different model or dataset would produce.

The design also clarifies why the proposed architecture is an empirical question. A rule score and a deep score can be combined in a single ranking function, but a combined output does not identify the contribution of either component. The no-Wide and deep-only conditions, train-only rule fitting, and shared evaluator are needed to distinguish a mechanism effect from differences in inputs, sampling, or optimization. Prior hybrid systems and basket studies support this component-oriented design, while retaining source-specific task assumptions [11] [12] [21]

The namespace policy has a second benefit. It makes an incomplete or incomparable source reproduction visible without treating a convenient public run as a replacement. It also keeps the controlled benchmark honest about its status as a constructed Vietnamese retail setting. This distinction is especially important for cold-item claims: graph methods learn from observed edges, and content signals can carry popularity structure, so neither sparse graph results nor available text alone identifies performance for a strict zero-edge item [24] [35]

This boundary also gives a practical interpretation of the current dataset decision. A catalog with 5,200 items and 5,000 users is not invalid merely because its interaction matrix is sparse. Sparse data can be appropriate for a controlled mechanism study when the target events, eligible-user count, item-coverage profile, and rule-support distribution are measured before the test is opened. The relevant question is whether enough users and items remain in each predeclared cohort to make the estimand stable and interpretable. The follow-up snapshot must consequently report the number of users with a novel positive, the number of strict zero-edge and sparse items, the number of transactions available for rule mining, and the exclusions caused by missing content or malformed timestamps. If those diagnostics show that the rule or cold-item estimand is underpopulated, the appropriate response is a versioned dataset amendment or a reduced-method ablation. It is not to change the generator, cohort definition, or metric after observing the test output.

The staged evidence policy is useful beyond this particular benchmark. It allows infrastructure and provenance to be validated on a public dataset without claiming that public-data behavior transfers to a constructed local setting. It also prevents a difficult source reproduction from being hidden by an easier surrogate. For a methods paper, this separation makes a negative or incomplete empirical outcome publishable as an explicit boundary rather than forcing every lane into a single success/failure label.

6.2 Limitations

First, the current public evidence is not an evaluation of the proposed model. It contains a deterministic sanity control and one learned reference family, with three BPR seeds but only one Pop run. Aggregate metrics are available, but paired per-user vectors, confidence intervals, and bootstrap estimates are not available in this lane. The descriptive direction should therefore not be treated as a general algorithm ranking.

Second, the strict source-native reproduction lane is closed as incomparable. This prevents a paper number from being transferred across a changed split, candidate policy, or evaluator, but it also means that the current manuscript cannot report a successful source-center reproduction. The follow-up must either resolve the exact source contract or retain the incomparable status.

Third, the v5 benchmark is controlled and semi-synthetic. It is useful for testing a declared mechanism under a reproducible schema, but its generated behavior may not represent the distribution, incentives, or repeat/explore balance of a live retail population. Next-basket work shows why repeat and novel item behavior should be separated rather than summarized by one aggregate quantity [20] [22]. The manuscript consequently treats the v5 counts as manifest declarations until the new snapshot lineage is admitted and does not describe the records as observed shopper logs.

Fourth, cold-item and content conclusions remain narrow. A strict zero-edge definition is not interchangeable with a rare-item threshold, and the denominators used in other cold-start studies do not automatically apply here. Content representations can be aligned to collaborative teachers and inherit their popularity structure; a text encoder can therefore be useful without settling the ranking question [28] [31].

Fifth, external validity is not established. A compatible public commerce dataset must preserve user, purchase, item-content, split, and candidate semantics. Shared authorship, platform, or benchmark lineage also limits the number of independent replications that can be claimed. These restrictions are particularly relevant for transfer methods whose downstream adaptation is part of the method rather than an optional implementation detail.

Finally, the current stage does not establish online behavior, revenue, latency gates, resource efficiency, or deployment readiness. The efficiency analysis is intentionally deferred until accuracy artifacts pass their integrity checks. These limitations are design constraints for the next experiment, not missing values to be filled by importing source-native metrics.

6.3 Follow-up decision criteria

The next empirical packet can support a model-effect conclusion only when the v5 snapshot, protocol, comparator registry, component artifacts, three seeds, per-user metrics, and bootstrap report are all sealed. The rule branch must be evaluated on its predeclared cohort, the cold analysis must retain separate denominators, and any external run must document deviations from the source contract. If any of these conditions fails, the corresponding conclusion stays unresolved. If the Hybrid contrast is null or negative, that outcome is reported directly and the paper’s interpretation is narrowed accordingly.

7 Conclusion

This paper defines a reproducible protocol for comparing retail recommenders under a fixed temporal, novel-purchase, full-catalog ranking task. Its central contribution is to bind data, split, candidate catalog, masking, evaluator, model artifacts, and uncertainty analysis into an auditable evidence chain. The completed public validation lane contains four descriptive Pop/BPR rows and shows that the chain can be executed and traced in a separate public namespace.

The evidence does not yet answer whether the proposed decoupled Wide-and-Deep Hybrid shows a positive contrast on the controlled Vietnamese benchmark. That question requires the separately audited v5 follow-up with component ablations, cold-item denominators, per-user statistics, and an explicit external-compatibility gate. Until those artifacts are available, the appropriate conclusion is a bounded methods contribution and a transparent empirical agenda. The next gate is therefore operationally precise: it must either produce a complete, null, or negative comparison under the frozen contract, or document why the relevant estimand remains unresolved. Both outcomes are more informative than importing a result from a different protocol.

Data Availability

The public validation uses a named public dataset under its provider terms. The controlled v5 data are a versioned research asset whose release and redistribution status must be checked independently; this manuscript does not assert unrestricted redistribution of row-level histories.

Ethics Statement

The reported work uses secondary public data and controlled/generated behavior; it does not recruit or interact with human participants. Any future use of private logs, identifiable records, surveys, or online experiments requires a new institutional ethics and data-protection determination.

Author Contributions

Author metadata and contribution roles are intentionally left blank in this venue-neutral version.

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